Some useful definitions and results

- If $X \sim \text{Beta}(\alpha, \beta)$, $\alpha > 0, \beta > 0$, $x \in (0,1)$, $p(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$. $E(X) = \alpha/(\alpha+\beta)$, $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.
- If $X \sim \text{Gamma}(\alpha, \beta)$, $\alpha, \beta > 0, x \geq 0$, $p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$. $E(x) = \alpha/\beta$, $Var(X) = \alpha/\beta^2$.
- For positive integers α, $\Gamma(\alpha) = (\alpha-1)!$
- If $X \sim \text{Binomial}(n, p)$, $p \in [0,1]$, $P(X = k) = \left(\begin{array}{c} n \\ k \end{array} \right) p^k (1-p)^{n-k}$, for $k = 0, \ldots, n$. $E(X) = np$, $Var(X) = np(1-p)$.
- If $X \sim \text{Poisson}(\lambda), \lambda > 0$, $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, \ldots,$. $E(X) = \lambda$, $Var(X) = \lambda$.
- If $X \sim N(\theta, \sigma^2)$, $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(x-\theta)^2)$, $E(X) = \theta, Var(X) = \sigma^2$.
- If $X \sim \text{Uniform}(a, b), b > a$, $p(x) = 1/(b-a)$, $E(X) = \frac{1}{2}(a+b)$, $Var(X) = \frac{1}{12}(b-a)^2$.

Some standard conjugate prior-likelihood pairs.

Likelihood	Prior	Posterior
$x \sim Binomial(n, \theta)$	$Beta(p, q)$	$Beta(p+x, q+n-x)$
$x_1, \ldots, x_n \sim Poisson(\theta)$	$Gamma(p, q)$	$Gamma(p + \sum_{i=1}^n x_i, q+n)$
$x_1, \ldots, x_n \sim N(\theta, \tau^{-1})$, ($\tau$ known)	$N(b, c^{-1})$	$N(\frac{cb+n\tau\bar{x}}{c+n\tau}, \frac{1}{c+n\tau})$
$x_1, \ldots, x_n \sim Gamma(k, \theta)$ (k known)	$Gamma(p, q)$,	$Ga(p+nk, q + \sum_{i=1}^n x_i)$
$x_1, \ldots, x_n \sim Geometric(\theta)$	$Beta(p, q)$	$Beta(p+n, q + \sum_{i=1}^n x_i - n)$
$x \sim NegBinomial(r, \theta)$	$Beta(p, q)$	$Beta(p+r, q+x)$